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# Using machine learning to improve survival prediction after heart transplantation

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## Abstract

**Background:** This study investigates the use of modern machine learning (ML) techniques to improve prediction of survival after orthotopic heart transplantation (OHT).

**Methods:** Retrospective study of adult patients undergoing primary, isolated OHT between 2000 and 2019 as identified in the United Network for Organ Sharing (UNOS) registry. The primary outcome was 1-year post-transplant survival. Patients were randomly divided into training (80%) and validation (20%) sets. Dimensionality reduction and data re-sampling were employed during training. Multiple machine learning algorithms were combined into a final ensemble ML model. The discriminatory capability was assessed using the area under receiver-operating-characteristic curve (AUROC), net reclassification index (NRI), and decision curve analysis (DCA).

**Results:** A total of 33,657 OHT patients were evaluated. One-year mortality was 11% ($n = 3738$). In the validation cohort, the AUROC of singular logistic regression was 0.649 (95% CI, 0.628–0.670) compared to 0.691 (95% CI, 0.671–0.711) with random forest, 0.691 (95% CI, 0.671–0.712) with deep neural network, and 0.653 (95% CI, 0.632–0.674) with Adaboost. A final ensemble ML model was created that demonstrated the greatest improvement in AUROC: 0.764 (95% CI, 0.745–0.782) ($p < .001$). The ensemble ML model improved predictive performance by 72.9% ±3.8% ($p < .001$) as assessed by NRI compared to logistic regression. DCA showed the final ensemble method improved risk prediction across the entire spectrum of predicted risk as compared to all other models ($p < .001$).

**Conclusions:** Modern ML techniques can improve risk prediction in OHT compared to traditional approaches. This may have important implications in patient selection, programmatic evaluation, allocation policy, and patient counseling and prognostication.

KEYWORDS

complications, heart failure, mortality/survival, transplantation

**Abbreviations:** AUROC, area under receiver-operating-characteristic curve; DCA, decision curve analysis; DRI, donor risk index; IHTSA, international heart transplant survival algorithm; IMPACT, index for mortality prediction after cardiac transplantation; ML, machine learning; NRI, net reclassification index; OHT, orthotopic heart transplantation; RSS, risk stratification score; STS, Society of Thoracic Surgeons; UNOS, United Network for Organ Sharing.

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## 1 | INTRODUCTION

The clinical management of end-stage heart disease is a continually evolving practice that has changed drastically over the last decade. Survival after orthotopic heart transplantation (OHT) has continued to improve.[1] Despite improving longevity after transplant, donor organ supply remains inadequate to meet demand,[2] and transplant programs face increased public and private scrutiny of their outcomes.[3] Simultaneously, technologic innovations in mechanical circulatory support platforms have demonstrated parallel improvement in clinical outcomes,[4–6] thus increasing the potential alternative viable treatment options for heart failure patients. Therefore, an accurate prognostic model using preoperative data for individualized donor and recipient selection would be of profound clinical utility in OHT.

Prior risk models for predicting survival after OHT have displayed only modest discriminatory capability. Examples of such algorithms include The Donor Risk Index (DRI),[7] Risk Stratification Score (RSS),[8] Index for Mortality Prediction After Cardiac Transplantation (IMPACT),[9] and International Heart Transplant Survival Algorithm (IHTSA).[10] With increasing interest in the application of machine learning (ML) to predictive analytics in clinical medicine,[11] we aimed to evaluate whether modern ML techniques could improve risk prediction in OHT.

## 2 | MATERIALS AND METHODS

### 2.1 | Patient population

Because of the sensitive nature of the data collected for this study, requests to access the data set from qualified researchers trained in human subject confidentiality protocols may be sent to the United Network for Organ Sharing (UNOS). We conducted a retrospective review of prospectively collected data in the UNOS database. The database was queried for all patients that underwent orthotopic heart transplantation (OHT) between 2000 and 2019. Patients were excluded if they underwent transplantation before the year 2000, were less than 18 years old, had a history of prior heart transplantation, underwent multiorgan transplant, or had incomplete survival status at one year. This retrospective analysis of deidentified data was deemed except from Institutional Review Board approval and patient consent was not required.

### 2.2 | Training and validation cohorts

Patients were randomly split into training (80%) and validation (20%) cohorts, ensuring equal distribution of the primary outcome in each cohort. This method of stratifying by primary outcome before randomly assigning patients to a cohort ensures an equal distribution of mortality between training and validation datasets to avoid biasing final model performance. The training cohort was used for feature selection, dimensionality reduction, and machine learning model development, keeping the validation cohort entirely separate and unseen until assessment of the final model performance.

### 2.3 | Data preparation and feature selection

All variables in the UNOS database available for the OHT patients were manually reviewed by two independent clinicians ($N = 525$ variables). Variables were excluded if they were redundant, free text, or would not be available in the preoperative setting. Variables with more than 20% missing data were also excluded. The distribution of data for each remaining categorical variable were again manually reviewed and grouped into clinically meaningful categories for each variable by two independent physicians. This step decreases data sparsity by grouping low incidence characteristics into fewer, clinically meaningful categories. Missing continuous variable data were imputed using feature median and missing categorical data were imputed with the feature mode. Continuous variables were standardized to have a mean of zero and standard deviation of one. Categorical variables were one-hot encoded ensuring no linear dependencies between columns.

### 2.4 | Feature importance and dimensionality reduction

To increase the interpretability of the model, we first performed a univariate estimate of feature importance. A random forest classifier was trained with 10-fold cross-validation repeated three times. Univariate feature importance was estimated using the mean decrease in accuracy. Three separate feature importance models were developed first with continuous variables only ($N = 37$), then with categorical variables ($N = 195$), and then with all the variables ($N = 232$). This was done to account for the known tendency for random forest classifiers to more heavily weight continuous variables in feature importance estimates. The 20 most important features from each model were combined into a single data set, consisting of 47 variables after excluding duplicates. This variable set was subsequently used to develop the machine learning algorithms. Overall, this methodology of feature selection allows us to combine the best of both manual filtering based on clinical acumen and automated methods using machine learning techniques. Decreasing the input space used for the final machine learning model is a critical step that has been shown to improve overall model performance by decreasing tendency to overfit and increasing training efficiency.[12]

### 2.5 | Training prediction models

An ensemble approach was employed to create a more stable and reliable model resistant to outliers. Four different types of algorithms were used: deep neural network, logistic regression, AdaBoost, and

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random forest. We employed over-sampling (SVM Smote) and under-sampling (Repeated Edited Nearest Neighbors) of the training data to better balance the primary outcome within the data set. For each type of algorithm, 100 different models were trained using varying degrees of data re-sampling to produce variability in each model's underlying training data. The resulting 400 algorithms were subsequently combined into the final ensemble prognostic model.

### 2.6 | Validation and comparison

Once the models were trained, discriminatory capability was assessed using the previously unseen validation data. The performance of the full ensemble model (400 algorithms) was compared to that of each type of algorithm individually (100 algorithms each), as well as a single logistic regression on its own. Model capability was assessed using area under receiver-operating-characteristic curve (AUROC), net reclassification index (NRI), and decision curve analysis (DCA). Calibration of the model was evaluated using visual plots of predicted risk based on the training cohort versus observed risk in the validation cohort stratified by decile of risk. Two-sided $p$ value of less than .05 was considered significant for all comparisons. All models were trained in python using Keras with Tensorflow.[13] Performance outcome comparisons were conducted with Stata (StataCorp. 2015, Stata Statistical Software: Release 14; StataCorp LP).

## 3 | RESULTS

### 3.1 | Study cohort

Of the 33,657 patients included in the study, 3728 (11%) experienced the primary outcome of death within one year (Figure 1). The majority of study participants were male (75%) with an average age of 52.8 ± 12.4 years. Balancing by primary outcome, patients were randomly split into 80% training cohort and 20% validation cohort. Patient characteristics were similar between cohorts, including age, sex, etiology of heart failure, end-organ function, pre-transplant mechanical circulatory support, and days spent on waiting list (Table 1).

### 3.2 | Feature importance and dimensionality reduction

After manual and automated variable filtering, the relative importance of each feature in predicting the primary outcome was assessed using a random forest classifier. Continuous and categorical variables were assessed both separately and together to reduce biases (Figure S1). In the combined assessment, the most important features included total bilirubin, mechanical ventilation, no ventricular assist device at transplant, serum creatinine, donor age, recipient height, donor cerebrovascular mechanism of death, prior cardiac surgery, and Karnofsky functional status. The top 20 most important features from each individual assessment were then combined excluding duplicates, resulting in 47 training variables that were used for subsequent model training (Table S1).

### 3.3 | Model training and performance comparison

A total of 400 algorithms were trained using varying subsets of training data based on randomly stratified levels of over and under resampling of the training data set (Figure 2). The final ensemble ML model contained all 400 underlying algorithms, while smaller sized ensemble models were also combined using the 100 iterations of each type of algorithm individually for comparison. The optimal model performance was the complete ensemble ML model (Figure 3), outperforming all other models with an AUROC of 0.764 (95% CI, 0.745–0.782) ($p < .001$). By comparison, the singular logistic regression model had an AUROC of 0.649 (95% CI, 0.628–0.670). Additionally, the final ensemble ML model demonstrated an improvement of 72.9% ± 3.8% ($p < .001$) in predictive performance as assessed by net reclassification index compared to logistic regression. The decision curve analysis showed the final ensemble method improved risk prediction across the entire spectrum of predicted risk as compared to all other models (Figure 4, $p < .001$). The final ensemble ML model was well-calibrated, with the majority of observed risk in the validation cohort falling within the range of predicted risk based on the training cohort after stratifying into deciles of risk (Figure S2).

## 4 | CONCLUSIONS

Prognostication of clinical outcomes after OHT has profound importance in patient selection and organ allocation. The present study demonstrates the potential utility of employing modern ML techniques to improve prognostic model performance at an individual patient level. A final ensemble ML model using only preoperative variables outperformed all other comparison algorithms within our study in predicting 1-year survival, achieving improved performance by a variety of metrics including AUROC, net reclassification index, and decision curve analysis. In one recent study, the performance of established OHT algorithms as assessed by AUROC remained only modest: IHTSA 0.65, DRI 0.56, RSS 0.61, and IMPACT 0.61.[10] Meanwhile, our final ensemble ML model was able to achieve an AUROC of 0.76, representing a significant improvement in overall accuracy which drastically increases the ability of clinicians to use such an algorithm in counseling patients and clinical decision making.

Moreover, this improvement in prognostication has the potential to advance how organs are allocated. Currently, heart allocation is determined predominately by the medical need of the recipient, compatibility of blood type, and physical proximity between the recipient and the donor. While these variables capture many of the important elements of a successful transplant (limiting ischemia time, minimizing risk for rejection, etc.) our machine learning model could

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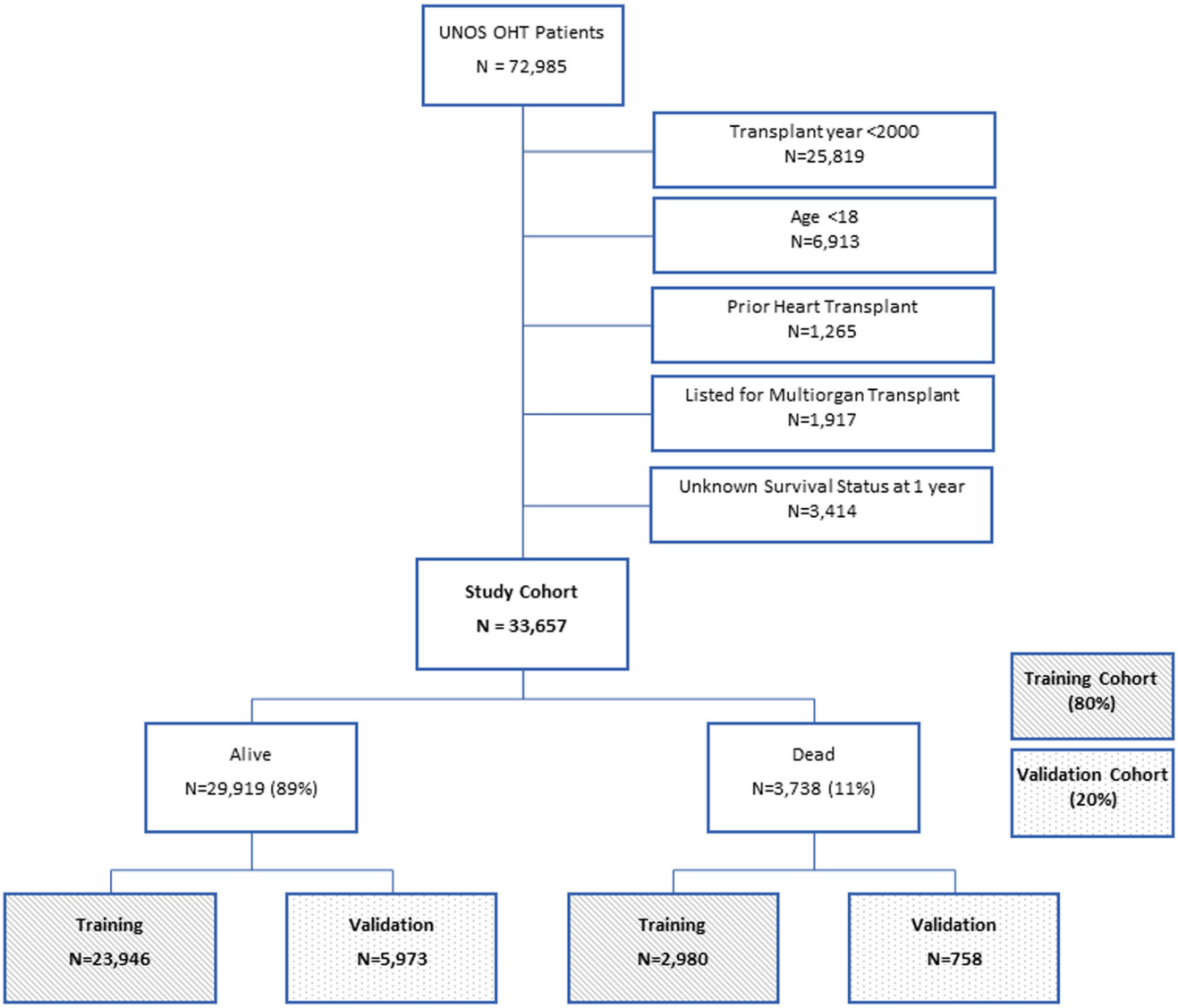


**FIGURE 1** Study cohort

potentially provide more nuanced insight into which recipient is expected to have the best outcome for a specific heart. Every time an organ becomes available, the algorithm could be run to predict each eligible transplant candidate's likelihood of one-year survival if they were to be chosen as the recipient. Such a personalized model could potentially improve outcomes by selecting the optimal donor-recipient pairing.

Cardiac surgery as a field has historically been an early adopter of clinical prognostic models,[14] most notably the widely used Society of Thoracic Surgeons (STS) Short-Term Risk Calculators.[15] However, predicting 1-year mortality after OHT has remained a persistent challenge. Early OHT risk models incorporated a select number of variables with only modest overall performance.[7,8,16] More recent models have added an increasing number of variables into more robust models, such as the IMPACT[9] and IHTSA,[10] but have been able to achieve only slight improvements in discriminatory performance. In part, this may relate to the challenges in capturing all granular and potentially predictive elements of post-transplant survival in a multicenter registry. For example, factors such as anti-rejection medication compliance are not assessed but can have important implications in survival following transplant. Also, there is a trade-off in assessing longer term outcomes, such that the event rate will be higher but the impact of preoperative risk factors on that outcome will likely diminish as longer-term factors weigh more heavily into outcome prediction.

Machine learning techniques have demonstrated clinical utility in a number of different fields.[17–21] Within OHT, the IHTSA score itself employs an artificial neural network approach, and has consistently demonstrated some of the highest discriminatory values out of all current models in recent studies.[22] Moreover, a recent study recalibrated both the IMPACT and IHTSA models to use only the same subset of variables and found the deep learning approach was superior.[23] The recent Trees of Predictors model is also an innovative approach that identifies clusters of patients with similar

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**TABLE 1** Patient characteristics of training and testing cohorts

| | Training ($n$ = 26,926) | % or *SD* | Validation ($n$ = 6731) | % or *SD* | *p* value |
|---|---|---|---|---|---|
| Recipient | | | | | |
| Age (years) | 52.8 | 12.5 | 52.8 | 12.3 | .873 |
| Female | 6683 | 25% | 1706 | 25% | .378 |
| Height (cm) | 173.9 | 9.8 | 173.7 | 9.9 | .099 |
| BMI (kg/m$^2$) | 27.0 | 4.8 | 27.1 | 4.8 | .453 |
| White | 18819 | 70% | 4723 | 70% | .666 |
| Cardiomyopathy | | | | | |
| Nonischemic | 12815 | 48% | 3239 | 48% | .445 |
| Ischemic | 10986 | 41% | 2703 | 40% | .339 |
| Congenital | 715 | 3% | 196 | 3% | .257 |
| Restrictive | 670 | 2% | 171 | 3% | .793 |
| Valvular | 509 | 2% | 130 | 2% | .842 |
| Hypertrophic | 580 | 2% | 152 | 2% | .607 |
| Prior cardiac surgery | 12012 | 45% | 2975 | 44% | .546 |
| History of dialysis | 596 | 2% | 169 | 3% | .144 |
| Karnofsky functional status >70% | 6748 | 25% | 1704 | 25% | .671 |
| Support: Ventilator | 506 | 2% | 115 | 2% | .389 |
| Support: ICU | 7789 | 29% | 1967 | 29% | .631 |
| Support: Inotropes | 11037 | 41% | 2694 | 40% | .149 |
| Support: ECMO | 127 | 0% | 38 | 1% | .329 |
| Support: IABP | 1576 | 6% | 379 | 6% | .503 |
| Mechanical Circulatory Support | | | | | |
| LVAD | 7465 | 28% | 1908 | 28% | .309 |
| RVAD | 50 | 0% | 9 | 0% | .419 |
| TAH | 208 | 1% | 43 | 1% | .268 |
| LVAD + RVAD | 599 | 2% | 149 | 2% | 1.000 |
| Total bilirubin | 1.1 | 2.0 | 1.1 | 1.6 | .271 |
| Creatinine | 1.3 | 0.6 | 1.3 | 0.6 | .713 |
| Total days on waiting list | 224.0 | 369.9 | 225.1 | 369.2 | .836 |
| Donor | | | | | |
| Age (years) | 31.8 | 11.9 | 31.9 | 12.1 | .413 |
| Female | 7830 | 29% | 1986 | 30% | .491 |
| Height (cm) | 174.3 | 9.6 | 174.1 | 9.7 | .290 |
| BMI (kg/m$^2$) | 26.9 | 5.6 | 27.0 | 5.8 | .336 |
| Mechanism of death | | | | | |
| Trauma | 14513 | 54% | 3634 | 54% | .902 |
| Cerebrovascular | 6859 | 25% | 1741 | 26% | .512 |
| Drug overdose | 1892 | 7% | 456 | 7% | .487 |

(Continues)

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**TABLE 1** (Continued)

| | Training (*n* = 26,926) | % or *SD* | Validation (*n* = 6731) | % or *SD* | *p* value |
|---|---|---|---|---|---|
| Other | 3662 | 14% | 900 | 13% | .633 |
| Blood infection | 2081 | 8% | 486 | 7% | .166 |
| Other | | | | | |
| Sex matched | 19891 | 66% | 4987 | 74% | .721 |
| Ischemic time (h) | 3.2 | 1.0 | 3.2 | 1.0 | .493 |
| Blood type incompatible | 3931 | 15% | 966 | 14% | .615 |
| Transplant year 2000–2010 | 13149 | 49% | 3365 | 50% | .091 |

Abbreviations: BMI, body mass index; ECMO, extracorporeal membrane oxygenation; IABP, intra-aortic balloon pump; ICU, intensive care unit; LVAD, left ventricular assist device; MCS, mechanical circulatory support; RVAD, right ventricular assist device; TAH, total artificial heart.

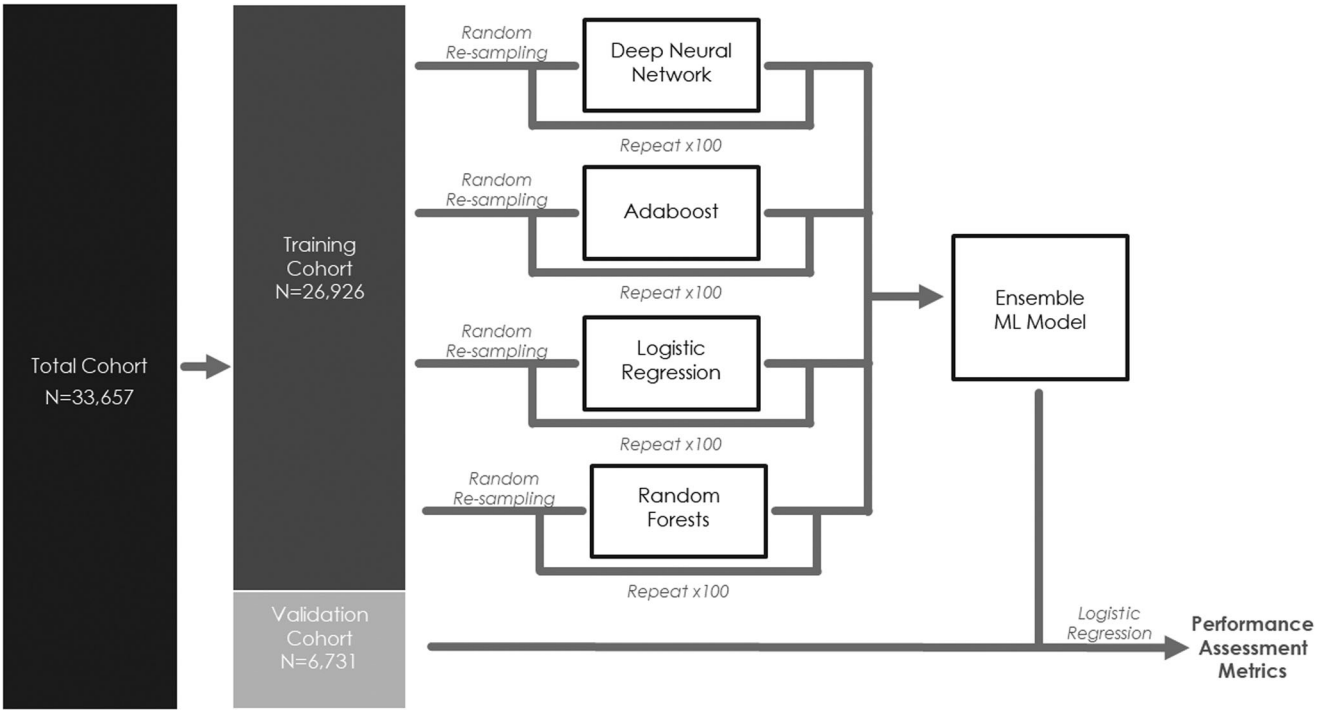


**FIGURE 2** Schematic of final ensemble machine learning (ML) model training and internal validation methodology

characteristics and develops machine learning predictive models specifically for each cluster.[24] The success of this approach demonstrates the potential for developing very individualized prognostic scores, at the risk of overfitting the model to specific retrospective cohorts that may not translate to prospective clinical practice.

The final ensemble ML model we developed in the present study is an example of using both clinical acumen and automated machine learning to develop a robust model from a large clinical registry. The statistical adage of "garbage in produces garbage out" remains especially true for machine learning approaches.[25] It is particularly relevant for black-box algorithms when used clinically, as there is low interpretability for clinicians in terms of how the algorithm arrives at its final prognosis. Moreover, registry data is particularly prone to reporting inaccuracies and missing data, resulting in poor prognostic ability if machine learning approaches are applied without sufficient data preparation.[26] Our approach was to combine both expert clinician manual review of the variables with automated feature selection techniques to arrive at the final set of variables. While time-consuming, we believe this collaborative approach is necessary to derive utility from registry-level data. Moreover, while computationally more expensive, the ensemble machine learning approach allows for the integration of multiple types of algorithms into one cohesive model, which has been suggested to produce a more robust final product.[27]

While this is not the first study to employ machine learning techniques for OHT prediction, it describes the use of more robust feature selection techniques and the development of a larger scale ensemble ML

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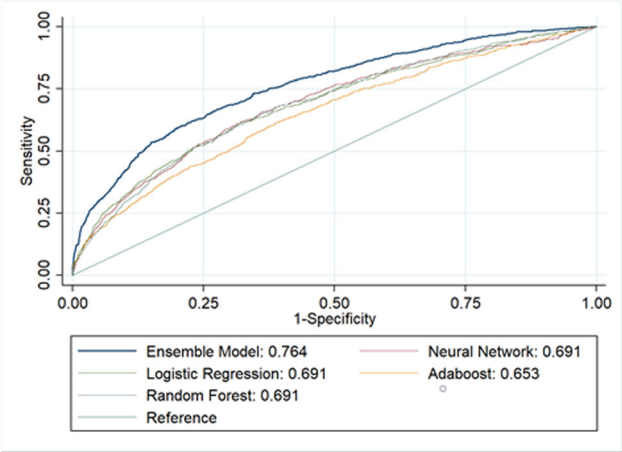


| Model Type | C-Statistic | 95% Confidence Interval | p-value |
|---|---|---|---|
| Full Ensemble Model | 0.764 | (0.745, 0.782) | *reference* |
| Neural Network Ensemble | 0.691 | (0.671, 0.712) | <0.0001 |
| Logistic Regression Ensemble | 0.691 | (0.670, 0.712) | <0.0001 |
| Adaboost Ensemble | 0.653 | (0.632, 0.674) | <0.0001 |
| Random Forest Ensemble | 0.691 | (0.671, 0.711) | <0.0001 |
| Logistic Regression (singular) | 0.649 | (0.628, 0.670) | <0.0001 |

**FIGURE 3** Area under receiver-operating-characteristic curve comparison of each model's prognostic ability

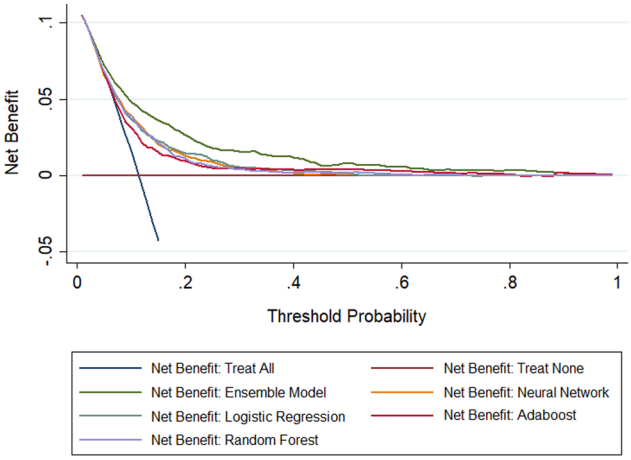


**FIGURE 4** Decision curve analysis demonstrating improved risk prediction of the full ensemble machine learning (ML) model across the entire spectrum of predicted risk scores

model than has been previously reported. This example of applying modern techniques may help to overcome the registry-level data limitations that have hindered prior studies.

This study has several limitations that need to be considered when interpreting the results. First, it is retrospective in nature and subject to all inherent limitations of such studies. Most notably, there have been a number of substantial changes in the allocation system and clinical management of OHT patients over the timeframe encapsulated by the study period. As such, there is associated bias as risk models including the one developed in the current study cannot account for individual provider or transplant program decision-making. Second, the UNOS database, similar to other multicenter registries, has a number of limitations including variability in data reporting and quality. As such, assumptions are made for missing data that may introduce bias and there may be clinically important variables not captured in the available data set. Finally, while we created a randomly selected validation cohort at the outset of the study, an independent validation cohort separate from the UNOS database was not available for testing. Further study is warranted on independent, prospective data not present in the current data set to provide more comprehensive validation testing of the final model.

In conclusion, an ensemble ML model was able to achieve greater predictive performance as compared to individual ML models and logistic regression in predicting survival after OHT. This analysis demonstrates the potential of modern ML techniques in risk prediction for OHT. These approaches may have important implications in patient selection, programmatic evaluation, policy-making, and patient counseling in OHT.

## DISCLOSURES

The data reported here have been supplied by the United Network for Organ Sharing as the contractor for the Organ Procurement and Transplantation Network. The interpretation and reporting of these data are the responsibility of the authors and in no way should be seen as an official policy of or interpretation by the OPTN or the U.S. Government. Arman Kilic is on the Medical Advisory Board for Medtronic, Inc. and a consultant for Abiomed. Igor Gosev is a consultant for Abbott, Inc. Sunil Prasad is on the Scientific Advisory Board for Abbott, Inc.

## MEETING PRESENTATION

American Heart Association Scientific Sessions 2020 (virtual meeting).

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## SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

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